

Fruit & Vegetable Department

Data-Driven Improvement Project Plan

Foodland Wudinna, SA

Prepared: March 2026

Owner: Fábio Henrique

Time commitment: 4–6 hours per week

A practical, one-person project plan to reduce waste, improve ordering accuracy, and build simple reporting tools for the Fruit & Veg department.

1. Why This Project

The Fruit & Vegetable department is the store's freshness signal — it's the first thing customers see. But the data shows avoidable losses are growing faster than revenue is declining. Without intervention, waste will continue to erode the department's profitability.

The problem in three numbers

Metric	Measured Period	Full-Year Projection
Department revenue (2026)	\$2,430/day (66 days)	\$636K/year
Department gross profit (2026)	\$924/day	\$242K/year
Avoidable waste (REDUCED FV)	3.6 kg/day and growing	\$4,300–\$8,600/year (est.)

Revenue projections are based on 308 trading days per year, adjusted for seasonal demand (summer peak 100%, winter 70%, shoulder months 80–90%). The department generates an estimated \$636K in annual revenue and \$242K in gross profit.

The exact waste figure is uncertain because the POS system strips the original PLU from reduced items, making true cost calculation impossible. Estimates range from \$4,300 to \$8,600/year depending on the assumed original product value and bin-to-reduced ratio. One of this project's goals is to close that data gap with real numbers.

2. Industry Benchmarks

These benchmarks support the case for investing time in data-driven waste reduction. The numbers come from published research and industry reports.

Benchmark	Source	Value
Avg fresh fruit shrinkage	USDA Economic Research Service	12.6% of sales
Avg fresh vegetable shrinkage	USDA Economic Research Service	11.6% of sales
Profit lift from data-driven ordering	Smith Business Insight / Queen's Uni.	7–10% profit increase
Waste reduction with analytics	RELEX Solutions case studies	Up to 40% reduction
Margin improvement with demand forecasting	Shelf Engine (US grocery chain)	15.6% → 25.6% margin
Operational shrink (not theft)	NRF Survey 2018	64% of all shrink

Key takeaway

Most shrinkage (64%) comes from operational issues — ordering, markdowns, and stock rotation — not theft. Simple tools and better processes can recover 7–10% of profit without any new software investment.

3. Project Scope

This project is designed for one person working 4–6 hours per week. It focuses on simple, practical tools that can be built with Excel and Python (Jupyter notebooks). No new software purchases are required.

What this project IS

- A set of spreadsheets and notebooks that turn POS export data into actionable information.
- Simple processes (waste log, ordering checklist) that improve daily decisions.
- Weekly reports that track progress and highlight problems early.

What this project is NOT

- A full software system or app development project.
- A replacement for the POS system or Metcash ordering platform.
- Something that requires management to invest money — only staff time.

4. Products & Deliverables

Six products, delivered in three phases. Each product is a standalone tool that adds value immediately.

Phase 1 — Foundation (Weeks 1–4)

Goal: Start measuring what matters. No analysis is possible without clean data.

Product 1: Waste Log (Excel spreadsheet)

A daily log where staff record every item that is reduced or binned. Captures the original product name, estimated quantity, reason (overripe, damaged, unsold), and action taken (reduced, binned, donated).

- **Why:** The POS strips the original PLU from reduced items. Without this log, we cannot calculate real waste costs.
- **Effort:** 2 hours to build, then 5 minutes per day to fill in.
- **Format:** Excel workbook with a data entry sheet and a weekly summary pivot.

Product 2: Weekly Ordering Aid (Excel spreadsheet)

A spreadsheet that calculates suggested order quantities for the top 30 items based on the last 4 weeks of same-weekday sales. Covers both Tuesday and Friday delivery cycles.

- **Why:** Ordering by gut feel is the primary cause of waste. A 4-week moving average is simple and already proven to beat intuition.
- **Effort:** 4 hours to build, then 15 minutes per order day to update.
- **Format:** Excel with automatic refresh from POS CSV export.

Phase 2 — Visibility (Weeks 5–8)

Goal: Turn raw data into weekly reports that highlight wins, problems, and trends.

Product 3: Weekly Performance Report (Jupyter notebook → PDF)

An automated report generated each Monday from the previous week's POS export. Covers: revenue vs. target, top/bottom sellers, waste log summary, specials performance, and ordering accuracy (ordered vs. sold).

- **Why:** Without a regular report, improvements are invisible. This creates accountability and a feedback loop.
- **Effort:** 6 hours to build, then 20 minutes per week to run and review.
- **Format:** Jupyter notebook that exports to a clean PDF.

Product 4: Specials Performance Tracker (Excel spreadsheet)

A week-by-week log of which items were on Wednesday Special, what price they ran at, how many units sold, and what the GP impact was. Answers the question: did this special actually work?

- **Why:** Some specials drive traffic, others just erode margin. This tool identifies which is which.
- **Effort:** 3 hours to build, then 10 minutes per Wednesday to update.
- **Format:** Excel with a simple dashboard tab showing trends.

Phase 3 — Optimisation (Weeks 9–12)

Goal: Use 8+ weeks of clean data to build smarter tools and measure real financial impact.

Product 5: Waste Cost Calculator (Jupyter notebook)

Using 8 weeks of waste log data combined with POS product costs, this tool calculates the true dollar cost of waste by product. Replaces the current estimate (which relies on a fake POS cost) with real numbers.

- **Why:** This is the missing piece. We currently cannot prove the exact financial impact. This tool fixes that.

- **Effort:** 4 hours to build. Runs automatically from waste log + POS data.
- **Depends on:** 8 weeks of consistent waste log entries (Product 1).

Product 6: Monthly KPI Dashboard (Jupyter notebook → PDF)

A monthly summary that tracks the 5 core KPIs against targets. Shows trend lines, compares to baseline, and highlights whether interventions are working.

- **Why:** This is the evidence for management. It proves ROI and justifies continued investment of time.
- **Effort:** 4 hours to build. 30 minutes per month to run.
- **Format:** One-page visual dashboard exported as PDF.

5. Timeline

12-week plan, starting April 2026. All dates assume 4–6 hours per week.

Week	Milestone	Product	Hours
1	Build Waste Log template	P1: Waste Log	2
1–2	Build Ordering Aid (top 30 items)	P2: Ordering Aid	4
2	Start daily waste logging	—	0.5/day
3–4	Refine ordering aid, first ordering cycle test	P2 refinement	3
5–6	Build Weekly Performance Report	P3: Weekly Report	6
5–6	Build Specials Tracker	P4: Specials Tracker	3
7–8	Run reports, collect feedback, adjust	P3–P4 refinement	2
9–10	Build Waste Cost Calculator (needs 8 weeks of log data)	P5: Waste Calculator	4
11–12	Build Monthly KPI Dashboard	P6: KPI Dashboard	4
12	First full KPI review with management	All products	2

Total estimated effort

Approximately 55–65 hours over 12 weeks (4–6 hrs/week). After setup, ongoing maintenance is approximately 2–3 hours per week.

6. KPIs & Targets

Five metrics, measured weekly. Targets are set for the 12-week review point.

KPI	Current (2026)	Target (Week 12)	How Measured
REDUCED FV (kg/day)	3.6 kg	< 2.0 kg	Waste Log
Ordering accuracy (sold/ordered)	Unknown	> 85%	Ordering Aid vs POS
Waste cost (\$/day, real)	Unknown	Establish baseline	Waste Cost Calculator
Open-ring lines/day	2.1	< 0.5	POS export
Specials GP impact (\$/week)	Tracked, not targeted	Stable or improving	Specials Tracker

The first 8 weeks are about establishing baselines. We cannot set realistic cost targets until the Waste Cost Calculator (Product 5) gives us real numbers. The 12-week review is the point where we present actual vs. estimated savings.

7. Investment & Expected Return

Investment (cost)

This project requires no financial investment. The only cost is staff time:

- **Setup (weeks 1–12):** 4–6 hours per week = approximately 55–65 total hours.
- **Ongoing (after week 12):** 2–3 hours per week for maintenance, reporting, and ordering.

Department full-year financials (2026 projection)

Metric	Per Trading Day	Annual (308 days, seasonally adjusted)
Revenue	\$2,430	\$636K
Gross Profit (38.0%)	\$924	\$242K
Estimated avoidable waste	\$16–\$33	\$4,300–\$8,600

Seasonal adjustment: Jan–Mar at 100% (peak summer), Apr–May at 85%, Jun–Aug at 70% (winter), Sep–Oct at 80%, Nov–Dec at 90%. This gives 262 equivalent peak-season days across 308 raw trading days.

Expected return (conservative)

Based on industry benchmarks and the EDA findings:

Improvement Area	Mechanism	Conservative Estimate
Waste reduction (25–35%)	Better ordering + earlier rotation	\$1,100–\$3,000/year
Ordering accuracy	Reduced over-ordering on top 30 items	\$1,500–\$3,000/year
Specials optimisation	Drop underperforming, keep high-traffic	\$500–\$1,200/year
Open-ring elimination	Better data quality for all future analysis	Enables above
TOTAL ESTIMATED BENEFIT		\$3,100–\$7,200/year

Industry benchmarks indicate 7–10% GP improvement from data-driven ordering. Applied conservatively (3–5%) to this department's projected \$242K annual GP, that represents \$7,300–\$12,100 in potential annual benefit. Our bottom-up estimate of \$3,100–\$7,200 sits well within that range.

Return on time invested

At the conservative midpoint (\$5,150/year), every hour spent on this project returns approximately \$80–\$95 in recovered value. Even the low end of the range (\$3,100/year) returns \$48 per hour invested — well above the cost of the time.

8. Risks & Mitigations

Risk	Impact	Mitigation
Waste log not filled daily	Cannot calculate real costs	Keep it under 5 min/day. Laminated card at weigh station as reminder
POS data quality issues	Open-ring items hide true demand	PLU sticker project (Priority 2 in recommendations)
Time squeeze — busy weeks	Progress stalls	Phase 1 products work standalone. Even just the waste log has value
Results take time to show	Management loses patience	Weekly report shows activity; 12-week KPI review shows trend

9. What I Need

This project is designed to be self-sufficient. However, two things would help:

1. **Approval for 4–6 hours per week** to be allocated to this project alongside regular duties. This is the only investment required.
2. **Support for the open-ring elimination initiative** — assigning PLU codes to all items and reinforcing cashier training. This is a one-time effort that dramatically improves data quality for everything else.

Everything else — tools, data, analysis — I will build and maintain.

10. Next Steps

If approved, the project starts immediately:

When	Action
This week	Finalise Waste Log template and begin daily logging
Week 1–2	Build Ordering Aid for top 30 items; test on first Friday order
Week 4	Brief check-in: share first waste log data and ordering accuracy
Week 8	First Weekly Report delivered to management
Week 12	Full KPI review with real cost data and measured improvements

End of Project Plan